CLOSED-FORM SYMPLECTIC PACKINGS OF THE FOUR-BALL BY EQUAL BALLS

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ABSTRACT. For every integer $k \geq 10$ and every $\varepsilon \in (0,1)$ we construct explicit C^4 symplectic embeddings $f_0, \dots, f_{k-1} \colon \mathbb{B}^\circ(r_+) \to \mathbb{B}^\circ(R)$ in $\mathbb{R}^4$, where $r_+ > 1$ and $R = (k/(1-\varepsilon))^{1/4}$, whose images are pairwise disjoint, together with an explicit left inverse on the target ball. Unlike the Hamiltonian flow construction of Paper I, no ordinary differential equation is integrated anywhere: each f_i is a finite composition of elementary closed-form maps together with scalar roots of strictly monotone one-variable equations (one Kepler equation in the inverse strip map and generating-function angle/radial equations in two planar keyholes). The geometry adapts the degree-9 disc-bundle mechanism of Paper I—linear comparison slope $3\sqrt{k} - 9$ and three marked fibres mapped to the line at infinity of $\mathbb{CP}^2$—into direct Darboux charts, generating-function keyholes with explicit full-disc nesting and slit bounds, C^4 deck-equivariant toric twists, and a piecewise closed-form transfer across the three edges of the moment simplex of $\mathbb{CP}^2$. All parameter inequalities hold analytically for every $k \geq 10$ and $\varepsilon \in (0,1)$. This paper provides the complete mathematical proof for `program/solution_embedding.py`.

1. INTRODUCTION AND MAIN THEOREM

1.1. **Statement of the result.** Write $\mathbb{R}^4$ with coordinates $X = (q_1, q_2, p_1, p_2)$ and standard symplectic form $\omega_0 = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. Let $\mathbb{B}(r) = \{X \in \mathbb{R}^4 : |X| \leq r\}$ and $\mathbb{B}^\circ(r) = \{X \in \mathbb{R}^4 : |X| < r\}$ denote the closed and open Euclidean balls of radius r about the origin; in capacity units, $B(c) = \{\pi|z|^2 \leq c\}$ and $B^\circ(c) = \{\pi|z|^2 < c\}$, so $\mathbb{B}(1) = B(\pi)$ and $\mathbb{B}(R) = B(\pi R^2)$.

Paper I [Son26c] constructs, for every $k \geq 10$ and $\varepsilon \in (0,1)$, a C^∞ compactly supported Hamiltonian on $\mathbb{R} \times \mathbb{R}^4$ whose time-one flow packs k closed unit balls into $\mathbb{B}^\circ((k/(1-\varepsilon/4))^{1/4})$, verified in Lean 4. Several steps of that construction—the handle Moser isotopy and the deformation-to-the-normal-cone transport and return—are defined as time-one maps of stiff vector fields. When only direct evaluation of the packing embeddings and their inverses is required, every differential equation can be replaced by closed-form charts and generating functions.

Theorem 1.1 (Closed-form C^4 equal-ball packings)**.** *Let $k \geq 10$ be an integer and $\varepsilon \in (0,1)$. Put $\varepsilon_c = \min(\varepsilon, \frac{1}{100})$, $\eta = \varepsilon_c/(1+\sqrt{1-\varepsilon_c}) = 1 - \sqrt{1-\varepsilon_c}$, $m_s = \min(2 \cdot 10^{-6}, \frac{1}{8}\eta)$, and*

$$R_{\mathrm{req}} = \left(\frac{k}{1-\varepsilon}\right)^{1/4}, \quad R_c = \left(\frac{k}{1-\varepsilon_c}\right)^{1/4}, \quad R_s = (1-m_s)R_c, \quad r_+ = (1-m_s)\left(\frac{1-\eta/2}{1-\eta}\right)^{1/2}.$$

Then $r_+ > 1$ and $R_s < R_c \leq R_{\mathrm{req}}$, and with the explicit parameter tuple $\Pi^*(k, \varepsilon_c; \frac{1}{4})$ of Definition 7.1, the compositions

$$f_i = S_{\mathrm{out}} \circ \chi_\kappa \circ \mathcal{W} \circ \mathcal{B} \circ \Pi_{\mathrm{post}} \circ \Phi_g \circ \Psi_S \circ F^{(i)} \circ C_i \circ \mathrm{KR} \circ S_{\mathrm{in}} \qquad (0 \leq i \leq k-1)$$

defined in Sections 2–7 satisfy over $\mathbb{R}$:

(i) Each $f_i \colon \mathbb{B}^\circ(r_+) \to \mathbb{R}^4$ is C^4, injective, and symplectic ($f_i^*\omega_0 = \omega_0$), mapping $\mathbb{B}^\circ(r_+)$ onto an open subset of $\mathbb{B}^\circ(R_s) \subset \mathbb{B}^\circ(R_{\mathrm{req}})$.

Version 1.1; companion to [Son26c, Son26b, Son26a] in `https://github.com/s-JoL/Symplectic-Ball-Packing`.

(ii) The open images $f_0(\mathbb{B}^\circ(r_+)),\dots,f_{k-1}(\mathbb{B}^\circ(r_+))$ *are pairwise disjoint; in particular the restrictions* $f_i|_{\mathbb{B}(1)}$ *pack* k *closed unit balls disjointly into* $\mathbb{B}^\circ(R_{\mathrm{req}})$ *with occupied volume fraction* $k/R_{\mathrm{req}}^4 = 1-\varepsilon$.

(iii) Each stage factor of f_i *has an explicit inverse, and the composite inverse procedure* $\mathrm{Preimage}(j,Y)$ *of Section 7 satisfies* $\mathrm{Preimage}(i,f_i(X)) = (X,\mathrm{true})$ *for every* $X\in\mathbb{B}^\circ(r_+)$ *and* $0\le i<k$, *while certifying non-membership (*$|X|>1$ *or* $\mathrm{ok}=\mathrm{false}$*) whenever* $Y\notin f_j(\mathbb{B}(1))$.

1.2. **Architecture and correspondence with code.** Table 1 lists the stage maps of f_i in evaluation order together with their domains, symplectic forms, and modular source files in `program/embedding_src/` (bundled into `program/solution_embedding.py` by `program/build_embedding.py`). Regularity is C^ω on the algebraic/trigonometric charts, C^8 on the planar keyholes ($m=4$), C^7 on the four-dimensional keyhole suspensions ($S_4'\in C^7$ enters the conjugate drift), and C^4 on the marked-fibre twists ($m=2$ plateau bumps); hence $f_i\in C^4$.

Stage	Map	Domain → target & form	Section / file
1. Input scale	$S_{\mathrm{in}}(q_1,q_2,p_1,p_2)$	$\mathbb{B}^\circ(r_+)\to B^\circ(a)$, conformal $\pi^{-1}R_s^{-2}$	§7; `chain.py`
2. Strip map	$\mathrm{KR}(x,y,w)$	$B^\circ(a)\xrightarrow{\sim}\mathcal{C}_a=\{0<X<a-p,0<T<1\}$	§2.2; `keyhole.py`
3. Cell to flat	$C_i(X,T,w)$	$i<9$: group windows; $i\ge9$: fan stack	§4; `chain.py`
4–5. Comparator	$F^{(i)}$ ($F_2\circ F_1$ for $i\ge9$)	Keyhole $KS_{\tau(G)}$ + slit shift $c(G)$	§4; `comparator.py`
6–7. Twists	$\Psi=\Phi_g\circ\Psi_S$	Routes $\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3$ into free corridor	§5; `twists.py`
8. Post chart	$\Pi_{\mathrm{post}}(G,v,w)$	Deck-adapted Darboux chart on $\{p<h_K\}$	§6; `chain.py`
9. Corner band	$\mathcal{B}$ (`Band`)	Corner keyhole K_B along $\sigma=j(1-3p)$	§6; `transfer.py`
10. Simplex glue	$\mathcal{W}(\sigma,\psi,p,\vartheta)$	4 sector pieces + 3 bands $\to\mathbb{CP}^2\setminus L_\kappa$	§6; `transfer.py`
11. Ball chart	$S_{\mathrm{out}}\circ\chi_\kappa([Z])$	$\mathbb{CP}^2\setminus L_\kappa\xrightarrow{\sim}B^\circ(1)\to\mathbb{B}^\circ(R_s)$	§6; `transfer.py`

TABLE 1. Stage decomposition ('STAGES(i)' in `solution_embedding.py`). For a planar fibre coordinate $w=w_x+iw_y\in\mathbb{C}$, $p=\pi|w|^2$ and $\theta=\arg(w)/(2\pi)\in\mathbb{R}/\mathbb{Z}$ satisfy $\omega_V=dw_x\wedge dw_y=dp\wedge d\theta$.

2. Smooth primitives and the Knothe–Rosenblatt strip map

2.1. **Trigonometric steps, ramps, and plateau bumps.** For an integer $m\ge1$, let $c_0=2^{-2m}\binom{2m}{m}$ and $a_j=2(-1)^j\binom{2m}{m-j}/\binom{2m}{m}$ $(1\le j\le m)$, so that $\sin^{2m}(\pi u)=c_0\big(1+\sum_{j=1}^m a_j\cos(2\pi ju)\big)$. Define $S_m\colon\mathbb{R}\to[0,1]$ by $S_m(u)=0$ for $u\le0$, $S_m(u)=1$ for $u\ge1$, and

$$S_m(u)=u+\sum_{j=1}^m\frac{a_j}{2\pi j}\sin(2\pi ju)\qquad(0\le u\le1).$$

Lemma 2.1 (Properties of S_m, I_m, R_m, and $\beta_{w_0,w_1,m}$)**.** *(i)* $S_m'(u)=\sin^{2m}(\pi u)/c_0$ *on* $[0,1]$ *vanishes to order* $2m$ *at* $u=0,1$. *Hence* $S_m\in C^{2m}(\mathbb{R})$, $S_m'\in C^{2m-1}(\mathbb{R})$, S_m *is strictly increasing on* $[0,1]$ *with* $S_m(1-u)=1-S_m(u)$, *convex on* $[0,\frac12]$ *and concave on* $[\frac12,1]$; *in particular* $S_m(u)\le u$ *for all* $u\in[0,\frac12]$.

(ii) The antiderivative $I_m(u)=\int_0^{\mathrm{clamp}(u,0,1)}S_m(t)\,dt$ *equals* $\frac12u^2+\sum_{j=1}^m\frac{a_j}{(2\pi j)^2}(1-\cos(2\pi ju))$ *on* $[0,1]$ *with* $I_m(1)=\frac12$. *The convex ramp* $R_m(u)=I_m(\mathrm{clamp}(u,0,1))+\max(u-1,0)$

belongs to $C^{2m+1}(\mathbb{R})$*, equals* 0 *on* $(-\infty,0]$ *and* $u-\frac12$ *on* $[1,\infty)$*, and satisfies* $\max(0,u-\frac12)\le R_m(u)\le\max(0,u)$ *for all* $u\in\mathbb{R}$.

(iii) For $0<w_0<w_1$, $\beta_{w_0,w_1,m}(x)=1-S_m\big((|x|-w_0)/(w_1-w_0)\big)$ *belongs to* $C^{2m}(\mathbb{R})$ *(*$|x|$ *is not differentiated on* $(-w_0,w_0)$ *where* $\beta_{w_0,w_1,m}\equiv1$*, and all derivatives up to order* $2m$ *vanish at* $|x|=w_0,w_1$*), and its primitive* $\mathcal{B}(x)=\int_{-\infty}^{x}\beta_{w_0,w_1,m}(t)\,dt\in C^{2m+1}(\mathbb{R})$ *has total mass* $\mathcal{B}(\infty)=w_0+w_1$.

Proof. Since $\sin(\pi u)$ has simple zeros at $u=0,1$, $\sin^{2m}(\pi u)$ vanishes to order $2m$, giving $S_m\in C^{2m}(\mathbb{R})$. On $(0,\frac12)$, $S_m''(u)=(2m\pi/c_0)\sin^{2m-1}(\pi u)\cos(\pi u)>0$, so convexity with $S_m(0)=0$, $S_m(\frac12)=\frac12$ yields $S_m(u)\le u$ on $[0,\frac12]$ and concavity on $[\frac12,1]$ via $S_m(1-u)=1-S_m(u)$. Integrating the symmetry identity gives $I_m(1)=\frac12$, and $R_m'=S_m\in C^{2m}(\mathbb{R})$ gives $R_m\in C^{2m+1}(\mathbb{R})$ with supporting lines 0 and $u-\frac12$. Part (iii) follows from $w_0>0$ and $\int_{-\infty}^{\infty}\beta_{w_0,w_1,m}=2w_0+2(w_1-w_0)(1-I_m(1))=w_0+w_1$. □

2.2. **The fibrewise Knothe–Rosenblatt strip map.** Let $B^\circ(a)=\{(x,y,w)\in\mathbb{R}^2\times\mathbb{C}:\pi(x^2+y^2)+p<a\}$ $(p=\pi|w|^2)$ and $\mathcal{C}_a=\{(X,T,w)\in\mathbb{R}^2\times\mathbb{C}:p<a,\ 0<X<a-p,\ 0<T<1\}$.

Proposition 2.2 (Knothe–Rosenblatt ball-to-cell map)**.** *For* $(x,y,w)\in B^\circ(a)$*, set* $r(p)=\sqrt{(a-p)/\pi}>0$ *and* $c(x,p)=\sqrt{r(p)^2-x^2}>0$*, and define*

$$\begin{aligned}&X(x,p)=r^2\Big(\frac{\pi}{2}+\arcsin\frac{x}{r}+\frac{xc}{r^2}\Big)=\int_{-r}^{x}2\sqrt{r^2-s^2}\,ds,\\ &T(x,y,p)=\frac12+\frac{y}{2c},\qquad C(x,y,p)=\frac{X-xc}{\pi r^2}\,\frac{y}{2c},\qquad \mathrm{KR}(x,y,w)=\big(X,T,we^{2\pi iC}\big).\end{aligned}\tag{1}$$

Then $\mathrm{KR}\colon B^\circ(a)\to\mathcal{C}_a$ *is a real-analytic symplectomorphism satisfying* $\mathrm{KR}^*(dX\wedge dT+dw_x\wedge dw_y)=dx\wedge dy+dw_x\wedge dw_y$*. Its inverse on* $\mathcal{C}_a$ *solves Kepler's equation* $2\alpha+\sin(2\alpha)=2(X/r^2-\pi/2)$ *for the unique* $\alpha\in(-\frac{\pi}{2},\frac{\pi}{2})$*, sets* $x=r\sin\alpha$*,* $c=r\cos\alpha$*,* $y=(2T-1)c$*,* C *by* (1)*, and returns* $(x,y,we^{-2\pi iC})$.

Proof. For $(x,y,w)\in B^\circ(a)$ we have $p<a$, $|x|<r(p)$, and $|y|<c(x,p)$, so $r,c>0$ and $|x/r|<1$; hence KR is real-analytic on all of $B^\circ(a)$ (including $w=0$). At fixed $p<a$, $X(x,p)\in(0,\pi r^2)=(0,a-p)$ has $\partial_xX=2c>0$, and substituting $x=r\sin\alpha$ gives $X/r^2=\frac{\pi}{2}+\alpha+\frac12\sin(2\alpha)$, whose derivative $1+\cos(2\alpha)=2\cos^2\alpha>0$ on $(-\frac{\pi}{2},\frac{\pi}{2})$ makes KR^{-1} the exact inverse on $\mathcal{C}_a$.

For symplecticity it suffices by real-analyticity to check $w\neq0$. Put $w'=we^{2\pi iC}$, so $p'=p$ and $dw_x'\wedge dw_y'=dp\wedge d(\theta+C)=dp\wedge d\theta+dp\wedge d_{x,y}C$. Since $d_{x,y}X\wedge d_{x,y}T=(2c\,dx)\wedge(\frac{dy}{2c}+\partial_xT\,dx)=dx\wedge dy$, expanding $dX\wedge dT$ gives

$$dX\wedge dT+dw_x'\wedge dw_y'=dx\wedge dy+dp\wedge d\theta+dp\wedge\big(\partial_pX\,d_{x,y}T-\partial_pT\,d_{x,y}X+d_{x,y}C\big).$$

Differentiating $X=\int_{-r}^{x}2\sqrt{r^2-s^2}\,ds$ with respect to $r^2=(a-p)/\pi$ yields $\partial_pX=-\frac{1}{\pi}\int_{-r}^{x}(r^2-s^2)^{-1/2}ds=-\frac{X-xc}{\pi r^2}$, while $\partial_pc=-\frac{1}{2\pi c}$ gives $\partial_pT=\frac{y}{4\pi c^3}$. Since $C=(-\partial_pX)(T-\frac12)$ and $\partial_x\big(\frac{X-xc}{\pi r^2}\big)=\frac{2c-(c-x^2/c)}{\pi r^2}=\frac{1}{\pi c}$, we obtain

$$\partial_pX\,d_{x,y}T-\partial_pT\,d_{x,y}X+d_{x,y}C=-\frac{y}{4\pi c^3}(2c\,dx)-\frac{y}{2c}\Big(-\frac{dx}{\pi c}\Big)=0.$$ □

3. The smooth planar keyhole and full-disc bounds

3.1. **Generating-function construction.** Let $r_a > 0$, $d > 0$, $0 < W_1 < W_2 \le \frac{1}{6}$, $0 < \rho_1 < 1$, $\beta > 0$, $X_w > 0$, and $r_h > 0$ satisfy

$$(W_1 + W_2)(1 + \beta - \rho_1) = \beta, \tag{2a}$$

$$0 < r_h \le d \sin(2\pi W_1), \tag{2b}$$

$$\rho_1 \pi (d + r_a)^2 + \frac{1}{2} X_w < \pi (d - r_h)^2. \tag{2c}$$

Notice that (2c) implies $r_h < d$. Around the eccentric centre $c_* = (-d, 0) \in \mathbb{C}$, write $w = c_* + Re^{2\pi i\phi}$ with action $X = \pi R^2 \ge 0$ and angle $\phi \in [-\frac{1}{2}, \frac{1}{2}]$. Set $M(X) = X_w R_4(X/X_w)$, so $M'(X) = S_4(\min(X/X_w, 1)) \in [0, 1]$ and $X - \frac{1}{2}X_w \le M(X) \le X$ for all $X \ge 0$ (Lemma 2.1(ii)). Define the odd map Ψ on $[-\frac{1}{2}, \frac{1}{2}]$ by $\Psi(0) = 0$ and

$$\Psi'(\psi) = \rho_1 + (1 + \beta - \rho_1) S_4\Big(\frac{|\psi| - W_1}{W_2 - W_1}\Big) \qquad \big(|\psi| \le \tfrac{1}{2}\big).$$

Lemma 3.1 (Properties of $\Psi(\psi) - \psi$)**.** *Under* (2a), $\Psi(\pm\frac{1}{2}) = \pm\frac{1}{2}$, *and* $\Psi(\psi) - \psi$ *extends* 1*-periodically to an odd function in* $C^9(\mathbb{R})$. *On* $[0, \frac{1}{2}]$, $\Psi(\psi) - \psi \le 0$, *with* $\Psi(\psi) - \psi = -\beta(\frac{1}{2} - \psi)$ *on* $[W_2, \frac{1}{2}]$, $0 \le \psi - \Psi(\psi) \le (1 - \rho_1)\psi < W_2$ *on* $[0, W_2]$, *and* $|\Psi(\psi) - \psi| \le \frac{1}{2}\beta$ *everywhere.*

Proof. Using $I_4(1) = \frac{1}{2}$, $\Psi(\frac{1}{2}) - \frac{1}{2} = (\rho_1 - 1)W_1 + \frac{1}{2}(\rho_1 + \beta - 1)(W_2 - W_1) + \beta(\frac{1}{2} - W_2) = \frac{1}{2}\big(\beta - (W_1 + W_2)(1 + \beta - \rho_1)\big) = 0$ by (2a). Since $\Psi' \equiv 1 + \beta$ near $\pm\frac{1}{2}$, $\Psi(\psi) - \psi$ is affine of slope β across $\pm\frac{1}{2}$ (mod 1), hence C^9 on $\mathbb{R}/\mathbb{Z}$. On $[0, \frac{1}{2}]$, Ψ' is non-decreasing, so $\Psi(\psi) - \psi$ is convex and vanishes at $0, \frac{1}{2}$, giving $\Psi(\psi) - \psi \le 0$. At its unique minimum $\psi_* \in (W_1, W_2)$ (where $\Psi'(\psi_*) = 1$), integrating $\Psi' - 1 \ge \rho_1 - 1$ from 0 and $\Psi' - 1 \le \beta$ to $\frac{1}{2}$ gives the two upper bounds. □

Proposition 3.2 (Global C^8 symplectomorphism KS_τ)**.** *For* $\tau \in [0, 1]$, *define* $KS_\tau \colon \mathbb{C} \to \mathbb{C}$ *via the generating function* $S(X, \psi; \tau) = X\psi + \tau M(X)(\Psi(\psi) - \psi)$ *about* $c_* = (-d, 0)$*:*

$$\phi = \psi + \tau M'(X)\big(\Psi(\psi) - \psi\big), \qquad X_n = X + \tau M(X)\big(\Psi'(\psi) - 1\big), \tag{3}$$

sending $w = c_* + \sqrt{X/\pi}\, e^{2\pi i\phi}$ *to* $w' = c_* + \sqrt{X_n/\pi}\, e^{2\pi i\psi}$, *with generator* $g_\tau(w) = \partial_\tau S = M(X)(\Psi(\psi) - \psi)$. *Then* $KS_\tau \in C^8(\mathbb{C}, \mathbb{C})$ *is a global symplectomorphism of* $(\mathbb{C}, dw_x \wedge dw_y)$ *with explicit inverse,* $g_\tau \in C^9(\mathbb{C}, \mathbb{R})$, *and for any* C^k *(*$2 \le k \le 8$*) profile* $\tau = \tau(G) \in [0, 1]$, *the suspension* $(G, v, w) \mapsto \big(G, v - \tau'(G) g_{\tau(G)}(w), KS_{\tau(G)}(w)\big)$ *is a* C^{k-1} *symplectomorphism of* $(\mathbb{R}^2 \times \mathbb{C}, dG \wedge dv + dw_x \wedge dw_y)$.

Proof. Since $\partial_\psi \phi = 1 + \tau M'(X)(\Psi'(\psi) - 1) \ge (1 - \tau M') + \tau M' \rho_1 \ge \rho_1 > 0$ and $\partial_X X_n = 1 + \tau(\Psi'(\psi) - 1)M'(X) \ge \rho_1 > 0$ with $X_n(0) = 0$ and $\lim_{X\to\infty} X_n = \infty$, both scalar equations have unique global roots on $\mathbb{R}/\mathbb{Z}$ and $[0, \infty)$ (for $X \ge X_w$, $M(X) = X - \frac{1}{2}X_w$ gives the closed-form inverse $X = (X_n + \frac{1}{2}\tau(\Psi'(\psi) - 1)X_w)/(1 + \tau(\Psi'(\psi) - 1))$). As $X = \pi|w - c_*|^2 \to 0$, $S_4'(u) = O(u^8)$ implies $M'(X) = O(|w - c_*|^{18})$ and $M(X) = O(|w - c_*|^{20})$, so $KS_\tau(w) - w = O(|w - c_*|^{17})$ and $g_\tau(w) = O(|w - c_*|^{20})$ are C^8 and C^9 across $w = c_*$. Finally, $dS = \phi\, dX + X_n\, d\psi + g_\tau\, d\tau$ gives $dX_n \wedge d\psi - dX \wedge d\phi = -dg_\tau \wedge d\tau$, so substituting $\tau = \tau(G)$ yields $dG \wedge d(v - \tau' g_\tau) + dX_n \wedge d\psi = dG \wedge dv + dX \wedge d\phi$. □

3.2. **Full-disc quantitative nesting, hole clearance, and slit clearance.**

Theorem 3.3 (Quantitative keyhole bounds and the constant $8.4 > 2 + 2\pi$)**.** *Assume* (2). *Put* $x_0 = \pi r_h^2$, $\theta_0 = W_1$, *and*

$$\delta_{\mathrm{rel}} = \beta\pi(r_a + d)(r_a + 8.4\, d), \qquad \delta_{\mathrm{small}} = 4\pi^3 d(r_a + d) W_2^2.$$

(i) (Disc nesting for all $\tau \in [0,1]$.) For every $P_0 \in [0, \pi r_a^2]$, every $\tau \in [0,1]$, and every $w \in \bar{D}(P_0) = \{z \in \mathbb{C} : \pi|z|^2 \le P_0\}$, $w' = KS_\tau(w)$ satisfies

$$\pi|w'|^2 \le \max\Big(P_0 + \max\big(\tau\,\delta_{\mathrm{rel}},\ \delta_{\mathrm{small}}\big),\ \pi d^2\Big). \tag{4}$$

In particular, whenever $P_0 \ge \pi d^2$, $\pi|w'|^2 \le P_0 + \max(\tau\,\delta_{\mathrm{rel}}, \delta_{\mathrm{small}})$.

(ii) (Hole and slit clearance at $\tau = 1$.) For every $w \in \bar{D}(\pi r_a^2)$, $w' = KS_1(w)$ satisfies

$$\pi|w'|^2 > x_0 = \pi r_h^2 \qquad \text{and} \qquad |\theta(w')| > \theta_0 = W_1,$$

where $\theta(w') = \arg(w')/(2\pi) \in (-\frac{1}{2}, \frac{1}{2}]$; thus $KS_1(\bar{D}(\pi r_a^2))$ is disjoint from $\bar{D}(x_0)$ and from the closed sector $\{|\theta| \le \theta_0\} \cup [0, \infty)$.

Proof. Let $w \in \bar{D}(P_0) \subset \bar{D}(\pi r_a^2)$. Then $R = |w - c_*| \le |w| + d \le r_a + d$ and $X = \pi R^2 \le \pi(r_a + d)^2$. Write $w' = c_* + R_n e^{2\pi i\psi}$ ($X_n = \pi R_n^2$). By conjugation symmetry $\bar{w} \mapsto \overline{w'}$, assume $\psi \in [0, \frac{1}{2}]$; then Lemma 3.1 gives $0 \le \phi \le \psi \le \frac{1}{2}$. From $|c_* + re^{2\pi i\alpha}|^2 = d^2 + r^2 - 2dr\cos(2\pi\alpha)$ we have

$$\pi|w'|^2 - \pi|w|^2 = (X_n - X) + 2\pi d(R - R_n)\cos(2\pi\psi) + 2\pi dR\big(\cos(2\pi\phi) - \cos(2\pi\psi)\big). \tag{5}$$

Proof of (i). Split $[0, \frac{1}{2}]$ by the sign of $\Psi'(\psi) - 1$.

Case A: $\Psi'(\psi) \le 1$ (so $\psi \le W_2 \le \frac{1}{6}$, $\cos(2\pi\psi) \ge \frac{1}{2} > 0$, and $R_n \le R$). If $R + R_n < 2d\cos(2\pi\psi)$, then $2R_n \le R + R_n < 2d\cos(2\pi\psi)$, so $R_n < d\cos(2\pi\psi)$ and directly

$$|w'|^2 = d^2 + R_n\big(R_n - 2d\cos(2\pi\psi)\big) \le d^2 \implies \pi|w'|^2 \le \pi d^2.$$

If instead $R + R_n \ge 2d\cos(2\pi\psi)$, then $(X_n - X) + 2\pi d(R - R_n)\cos(2\pi\psi) = \pi(R - R_n)\big(2d\cos(2\pi\psi) - (R + R_n)\big) \le 0$. Moreover, since $0 \le \phi \le \psi \le W_2$, $\cos(2\pi\phi) - \cos(2\pi\psi) \le 1 - \cos(2\pi\psi) = 2\sin^2(\pi\psi) \le 2\pi^2\psi^2 \le 2\pi^2 W_2^2$. Hence (5) gives $\pi|w'|^2 - \pi|w|^2 \le 2\pi d(r_a + d)(2\pi^2 W_2^2) = 4\pi^3 d(r_a + d)W_2^2 = \delta_{\mathrm{small}}$.

Case B: $\Psi'(\psi) > 1$ (so $1 < \Psi'(\psi) \le 1 + \beta$ and $R_n \ge R$). First, $X_n - X = \tau M(X)(\Psi'(\psi) - 1) \le \tau\beta X \le \tau\beta\pi(r_a + d)^2$. Second, if $\cos(2\pi\psi) \ge 0$ then $2\pi d(R - R_n)\cos(2\pi\psi) \le 0$; if $\cos(2\pi\psi) < 0$ then $R_n - R = (X_n - X)/(\pi(R_n + R)) \le \tau\beta\pi R^2/(2\pi R) = \frac{1}{2}\tau\beta R$, so $2\pi d(R - R_n)\cos(2\pi\psi) \le 2\pi d(R_n - R) \le \tau\beta\pi d(r_a + d)$. Third, by Lemma 3.1, $0 \le \psi - \phi = \tau M'(X)(\psi - \Psi(\psi)) \le \frac{1}{2}\tau\beta$, so $\cos(2\pi\phi) - \cos(2\pi\psi) \le 2\pi(\psi - \phi) \le \pi\tau\beta$, giving $2\pi dR(\cos(2\pi\phi) - \cos(2\pi\psi)) \le 2\pi^2\tau\beta d(r_a + d) = (2\pi)\tau\beta\pi d(r_a + d)$. Summing the three terms in (5) yields

$$\pi|w'|^2 - \pi|w|^2 \le \tau\beta\pi(r_a + d)\big[(r_a + d) + d + 2\pi d\big] = \tau\beta\pi(r_a + d)\big[r_a + (2 + 2\pi)d\big] \le \tau\,\delta_{\mathrm{rel}},$$

since $2 + 2\pi = 8.283185\cdots < 8.4$. Combining Cases A and B proves (4).

Proof of (ii). Set $\tau = 1$. If $\psi \in [0, W_1]$, then $\Psi'(\psi) = \rho_1$ and $M(X) \ge X - \frac{1}{2}X_w$, so by (2c),

$$X_n = X - (1 - \rho_1)M(X) \le \rho_1 X + \tfrac{1}{2}(1 - \rho_1)X_w < \rho_1\pi(d + r_a)^2 + \tfrac{1}{2}X_w < \pi(d - r_h)^2.$$

Hence $R_n < d - r_h$, which places w' strictly in the left half-plane $\mathrm{Re}(w') = -d + R_n\cos(2\pi\psi) < -r_h < 0$; consequently $|w'| \ge |\,\mathrm{Re}(w')| > r_h$ ($\pi|w'|^2 > x_0$) and $|\theta(w')| > \frac{1}{4} > \frac{1}{6} \ge W_2 > W_1$. If instead $\psi \in (W_1, \frac{1}{2}]$, then for $\psi \ge \frac{1}{4}$ we again have $\mathrm{Re}(w') \le -d < -r_h$, whereas for $\psi \in (W_1, \frac{1}{4})$ the ray $\{c_* + te^{2\pi i\psi} : t \ge 0\}$ has perpendicular distance $d\sin(2\pi\psi) > d\sin(2\pi W_1) \ge r_h$ from the origin by (2b), so $|w'| > r_h$ for all $R_n \ge 0$; moreover, whenever $\mathrm{Re}(w') > 0$ (so $R_n\cos(2\pi\psi) > d$), $\tan(2\pi\theta(w')) = \frac{R_n\sin(2\pi\psi)}{R_n\cos(2\pi\psi) - d} > \tan(2\pi\psi) > \tan(2\pi W_1)$, whence $\theta(w') > \psi > W_1 = \theta_0$. □

4. Cell layout and the four-dimensional keyhole comparator

4.1. **The flat model, fundamental domain, and cell embeddings C_i.** Let $k \ge 10$ and $\eta \in (0, 1 - \sqrt{99/100}]$, and set $s = \sqrt{k} > 3$, $t_0 = 1 - \frac{1}{2}\eta$, $a = t_0/s$, $h = t_0/3$, $c_s = 3s - 9 > 0$, $g = 3(1 - t_0) = \frac{3}{2}\eta$, and $T = g/c_s$. On the flat chart $(G, v, w) \in \mathbb{R} \times \mathbb{R} \times \mathbb{C}$ with symplectic

form $\Omega_{\mathrm{fl}} = dG \wedge dv + dw_x \wedge dw_y$, define $P(p) = 3 - 9p$, $J(p) = 3sp - 3$, and $K(p) = c_s p$ for $p = \pi|w|^2 < \frac{1}{3}$. Notice the exact period identity

$$K(p) - J(p) = (3s - 9)p - (3sp - 3) = 3 - 9p = P(p) > 0, \tag{6}$$

and the deck group generated by the two C^ω symplectomorphisms

$$T_u(G, v, w) = \big(G + P(p),\, v,\, we^{2\pi i(9v)}\big), \qquad T_v(G, v, w) = (G, v + 1, w),$$

which commute modulo integer fibre turns ($T_u T_v = T_v T_u$ because $e^{2\pi i \cdot 9} = 1$) and have fundamental domain $\mathcal{F}_{\mathrm{fl}} = \{p < \frac{1}{3},\ J(p) \le G < K(p),\ 0 \le v < 1\}$. Define the level curves

$$O(G) = \frac{G + 3}{3s}, \quad E_{\mathrm{lin}}(G) = \frac{G}{c_s}, \quad P_{\mathrm{top}}(G) = \frac{ka - 3 - G}{k - 3s}, \quad \mathrm{slack}(G) = T\Big(1 + \frac{\min(G, 0)}{g}\Big).$$

Direct algebra gives the linear comparison identity of Paper I in level-G form:

$$O(G) - E_{\mathrm{lin}}(G) - P_{\mathrm{top}}(G) \equiv T \qquad (\forall\, G \in \mathbb{R}), \tag{7}$$

and $p < O(G) \iff G > J(p)$, while $p > E_{\mathrm{lin}}(G) \iff G < K(p)$.

Lemma 4.1 (Cell-to-flat embeddings C_i)**.** *For $0 \le i \le k - 1$, define $C_i \colon \mathcal{C}_a \to \mathbb{R} \times \mathbb{R} \times \mathbb{C}$ on $(X, T_{\mathrm{cell}}, w)$ ($p = \pi|w|^2 < a$, $0 < X < a - p$, $0 < T_{\mathrm{cell}} < 1$) by*

$$C_i(X, T_{\mathrm{cell}}, w) = \begin{cases} \big(J(p) + 9X,\ \frac{i-4+T_{\mathrm{cell}}}{9},\ we^{-2\pi i(3sT_{\mathrm{cell}}/9)}\big), & 0 \le i < 9, \\ \big(J(p) + i(a - p) + X,\ T_{\mathrm{cell}} + \frac{1}{2},\ we^{-2\pi i((3s-i)T_{\mathrm{cell}})}\big), & 9 \le i \le k - 1. \end{cases}$$

Then each C_i is a real-analytic symplectic embedding ($C_i^* \Omega_{\mathrm{fl}} = dX \wedge dT_{\mathrm{cell}} + \omega_V$). The nine group images $\mathcal{G}_i = C_i(\mathcal{C}_a)$ ($0 \le i < 9$) are pairwise disjoint in $\mathcal{F}_{\mathrm{fl}}$ (lying in disjoint v-windows $(\frac{i-4}{9}, \frac{i-3}{9}) \subset (-\frac{4}{9}, \frac{5}{9})$) and satisfy

$$J(p) < G < J(p) + 9(a - p) = (3s - 9)p + 9a - 3 < 3sa - 3 = -g \qquad (\forall\, (G, v, w) \in \mathcal{G}_i). \tag{8}$$

The stack images $\mathcal{S}_i = C_i(\mathcal{C}_a)$ ($9 \le i < k$) are pairwise disjoint (separated by the hyperplanes $A_i(p) = J(p) + i(a - p)$), satisfy $G > J(p) + 9(a - p)$, $G < J(0) + ka = st_0 - 3 =: G_{\max}$, and at each level G have fibre slice $p \le P_{\mathrm{top}}(G)$ (with $p < a$).

Proof. For $i < 9$, $d(J(p) + 9X) \wedge d(\frac{i-4+T_{\mathrm{cell}}}{9}) = (3s\, dp + 9\, dX) \wedge \frac{dT_{\mathrm{cell}}}{9} = dX \wedge dT_{\mathrm{cell}} + \frac{s}{3} dp \wedge dT_{\mathrm{cell}}$, which cancels $dp \wedge d(\theta - \frac{s}{3} T_{\mathrm{cell}}) - dp \wedge d\theta$; similarly for $i \ge 9$, $d(J(p) + i(a - p) + X) \wedge dT_{\mathrm{cell}} = dX \wedge dT_{\mathrm{cell}} + (3s - i) dp \wedge dT_{\mathrm{cell}}$. Since $3s - 9 > 0$ and $p < a$, $(3s - 9)p + 9a - 3 < (3s - 9)a + 9a - 3 = 3sa - 3 = 3t_0 - 3 = -g$. Finally, $G < J(p) + k(a - p) = -3 + ka - (k - 3s)p$ is equivalent to $p < P_{\mathrm{top}}(G)$. □

4.2. **The four-dimensional keyhole comparator $F = F_2 \circ F_1$.** Fix $0 < \lambda_2 < \lambda_a < 1$ and $\mu \in (0, 1)$ ($\lambda_a = \frac{9}{10}$, $\lambda_2 = \frac{3}{10}$, $\mu = \frac{1}{2}$), and put $w_G = \lambda_a g$ and $w_2 = \lambda_2 g$. Let K_C be a smooth keyhole on $\bar{D}(a)$ satisfying (2) with hole $x_0 = \pi r_h^2$ and slit half-width $\theta_0 = W_1$. Define

$$\tau(G) = S_4\Big(\frac{G + w_G}{w_G - w_2}\Big), \quad c(G) = \frac{w_2}{c_s} R_4\Big(\frac{G + w_2}{w_2}\Big), \quad \vartheta(\theta) = \theta - \tfrac{1}{2}\operatorname{sgn}(\theta)\ \ \big(\theta \in (-\tfrac{1}{2}, \tfrac{1}{2}] \setminus \{0\}\big).$$

Notice that $\vartheta(\arg(w)/(2\pi)) = \arg(-w)/(2\pi) \in (-\frac{1}{2}, \frac{1}{2})$ is real-analytic on the slit plane $\mathbb{C} \setminus [0, \infty)$ (in particular across the negative real axis $\theta = \pm\frac{1}{2}$!). Define $F_1(G, v, w) = \big(G, v - \tau'(G) g_{\tau(G)}(w), KS_{\tau(G)}(w)\big)$ and, on $\{c(G) = 0\} \cup \{(G, v, w_1) : w_1 \in \mathbb{C} \setminus [0, \infty),\ \pi|w_1|^2 + c(G) > 0\}$, define $F_2(G, v_1, w_1) = (G, v_1, w_1)$ when $c(G) = 0$ ($G \le -w_2$) and

$$F_2(G, v_1, w_1) = \Big(G,\ v_1 - c'(G)\vartheta(\theta_1),\ \sqrt{\frac{x_1 + c(G)}{\pi}}\, e^{2\pi i \theta_1}\Big) \qquad (x_1 = \pi|w_1|^2,\ \theta_1 = \arg(w_1)/(2\pi))$$

when $c(G) > 0$ ($G > -w_2$). Set $F = F_2 \circ F_1$, $F^{(i)} = \mathrm{id}$ for $0 \le i < 9$, and $F^{(i)} = F$ for $9 \le i < k$.

Theorem 4.2 (Domain, symplecticity, and stack-corridor separation of F)**.** *Suppose K_C satisfies* (2) *on $\bar{D}(a)$ together with the schedule inequalities:*

- *(H6)* $\max\big(S_4(u)\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}\big) \le (1-\mu)T\big(1-\lambda_a+u(\lambda_a-\lambda_2)\big)$ *for* $u\in[0,1]$*;*
- *(H7)* $\max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + \lambda_2 T R_4(u) \le (1-\mu)T\big(1-\lambda_2(1-u)\big)$ *for* $u\in[0,1]$*;*
- *(H8)* $\pi d^2 \le T(1-\mu-\frac{1}{2}\lambda_2)$ *and* $\pi d^2 + \frac{1}{2}\lambda_2 T + \mu T \le O(-w_G)$.

Then on the open stack union $\mathcal{S} = \bigcup_{i=9}^{k-1} \mathcal{S}_i$:

(i) For every $(G, v, w) \in \mathcal{S}$ with $G > -w_2$, $\tau(G) = 1$ and $w \in B^\circ(a)$, so by Theorem 3.3(ii) $w_1 = KS_1(w)$ satisfies $x_1 > x_0 > 0$ and $|\theta_1| > \theta_0 > 0$, avoiding $[0,\infty)$ by an open sector. As $G \downarrow -w_2$, $c(G)$ and $c'(G)$ vanish to order 9 while $|\vartheta(\theta_1)| < \frac{1}{2}$ and $x_1 > x_0$. Hence F is a C^7 symplectic embedding on an open neighbourhood of $\mathcal{S}$, with explicit C^7 inverse $F_1^{-1} \circ F_2^{-1}$.

(ii) Every image point $(G, v_2, w_2) \in F(\mathcal{S})$ ($p_2 = \pi|w_2|^2$) lies strictly inside a single fundamental domain $[J(p_2), K(p_2))$ of T_u and satisfies:

$$p_2 < h_K \qquad \text{(whenever } h_K \ge a \text{ and } h_K > \max_{G\in[-w_G, G_{\max}]} (O(G) - \mu\,\mathrm{slack}(G))\text{)}, \tag{9}$$

$$J(p_2) < G < K(p_2) - \min\big(w_2,\ c_s x_0 + \tfrac{1}{2}w_2\big) \le K(p_2) - \tfrac{1}{2}w_2, \tag{10}$$

$$G > J(p_2) + 9(a - p_2) \qquad \text{whenever } p_2 < a. \tag{11}$$

Consequently, the k sets $\mathcal{G}_0, \dots, \mathcal{G}_8, F(\mathcal{S}_9), \dots, F(\mathcal{S}_{k-1})$ are pairwise disjoint modulo the deck group $\langle T_u, T_v\rangle$, and all of them miss the corridor $\mathcal{K}_{\mathrm{free}} = \{p < \frac{1}{3} :\ K(p) - \frac{1}{2}w_2 \le G < K(p)\} \pmod{P(p)}$.

Proof. (i) F_1 is a global C^7 symplectomorphism by Proposition 3.2 ($\tau' \in C^7$). On $\{G > -w_2\}$, $d(x_1 + c(G)) \wedge d\theta_1 + dG \wedge d(v_1 - c'(G)\vartheta(\theta_1)) = dx_1 \wedge d\theta_1 + c'(G)dG \wedge d\theta_1 + dG \wedge dv_1 - c'(G)dG \wedge d\vartheta(\theta_1) = dG \wedge dv_1 + dx_1 \wedge d\theta_1$ since $\vartheta'(\theta_1) = 1$ off $\theta_1 = 0$. Because $\mathcal{S}_9, \dots, \mathcal{S}_{k-1}$ are pairwise disjoint and F is injective on their union, $F(\mathcal{S}_9), \dots, F(\mathcal{S}_{k-1})$ are pairwise disjoint.

(ii) If $G \le -w_G$, $F = \mathrm{id}$, so $p_2 = p < a \le h_K$, $J(p_2) + 9(a - p_2) < G \le -w_G < -w_2 < K(p_2) - w_2$. Now suppose $G \in (-w_G, G_{\max})$. Since $w \in B^\circ(\min(a, P_{\mathrm{top}}(G)))$, Theorem 3.3(i) gives $x_1 \le \max(P_{\mathrm{top}}(G) + \max(\tau(G)\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}), \pi d^2)$, and adding $c(G)$ gives $p_2 = x_1 + c(G)$. On $(-w_G, -w_2]$ ($c(G) = 0$), (H6) gives $P_{\mathrm{top}}(G) + \max(\tau\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) \le P_{\mathrm{top}}(G) + (1-\mu)\,\mathrm{slack}(G)$. Notice that for $G \le 0$, $G/c_s = T(G/g)$, so (7) gives the exact identity

$$O(G) - P_{\mathrm{top}}(G) = T + \frac{G}{c_s} = T\Big(1 + \frac{G}{g}\Big) = \mathrm{slack}(G) \qquad (G \le 0),$$

whereas for $G \ge 0$, $O(G) - P_{\mathrm{top}}(G) = T + E_{\mathrm{lin}}(G)$ with $\mathrm{slack}(G) = T$. Therefore:

- On $(-w_G, -w_2]$, (H6) gives $P_{\mathrm{top}}(G) + \max(\tau\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) \le O(G) - \mu\,\mathrm{slack}(G)$;
- On $[-w_2, 0]$, (H7) gives $P_{\mathrm{top}}(G) + \max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + c(G) \le O(G) - \mu\,\mathrm{slack}(G)$;
- On $[0, G_{\max}]$, $c(G) = \frac{G}{c_s} + \frac{1}{2}\lambda_2 T$, and (H7) at $u = 1$ gives $\max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + \frac{1}{2}\lambda_2 T \le (1-\mu)T$, so $P_{\mathrm{top}}(G) + \max(\delta_{\mathrm{rel}}, \delta_{\mathrm{small}}) + c(G) \le P_{\mathrm{top}}(G) + \frac{G}{c_s} + (1-\mu)T = O(G) - \mu T = O(G) - \mu\,\mathrm{slack}(G)$.

In the remaining $\pi d^2 + c(G)$ branch of (4): on $(-w_G, 0]$, $c(G) \le c(0) = \frac{1}{2}\lambda_2 T$ and $\mathrm{slack}(G) \le T$, so (H8) gives $\pi d^2 + c(G) + \mu\,\mathrm{slack}(G) \le \pi d^2 + \frac{1}{2}\lambda_2 T + \mu T \le O(-w_G) < O(G)$; on $[0, G_{\max}]$, $O(G) - \frac{G}{c_s} = P_{\mathrm{top}}(G) + T \ge T$, so (H8) gives $\pi d^2 + c(G) + \mu T = \pi d^2 + \frac{G}{c_s} + (\frac{1}{2}\lambda_2 + \mu)T \le \frac{G}{c_s} + T \le O(G)$. Thus $p_2 \le O(G) - \mu\,\mathrm{slack}(G) < O(G)$ holds for *all* $G \in (-w_G, G_{\max})$, which gives both (9) and $G > 3sp_2 - 3 = J(p_2)$!

For the upper bound (10): if $G \in (-w_G, -w_2]$, then $K(p_2) \ge 0$, so $K(p_2) - G \ge -G \ge w_2$. If $G \in (-w_2, G_{\max})$, Theorem 3.3(ii) gives $x_1 > x_0$, so $p_2 > x_0 + c(G)$; writing $u = (G + w_2)/w_2 > 0$

$(G = (u-1)w_2)$ and using $u - R_4(u) \le \lim_{t\to\infty}(t - R_4(t)) = \frac{1}{2}$, we have

$$K(p_2) - G > c_s x_0 + c_s c(G) - G = c_s x_0 + w_2\big(1 - (u - R_4(u))\big) \ge c_s x_0 + \tfrac{1}{2}w_2.$$

Finally, for $G > -w_G = -0.9g$ and any $p_2 < a$, (8) gives $J(p_2) + 9(a - p_2) < -g < -0.9g < G$, proving (11) and disjointness modulo $\langle T_u, T_v\rangle$. □

5. Marked-fibre twists and corridor routing

In the flat chart, put $u = G/P(p)$ $(P(p) = 3 - 9p > 0)$. The three straight marked fibres of the cubic transfer (Section 6) are $\mathcal{M}_1 = \{(u,v) \equiv (\frac{5}{6}, 0)\}$, $\mathcal{M}_2 = \{(u,v) \equiv (\frac{1}{3}, 0)\}$, and $\mathcal{M}_3 = \{(u,v) \equiv (0, \frac{1}{2})\}$ modulo $\mathbb{Z}^2$. Since the ball images fill (u,v) across (u_j, v_j), we interpose two closed-form C^4 deck-equivariant symplectomorphisms $\Psi = \Phi_g \circ \Psi_S$ that pull all three marked fibres back into the empty corridor $\mathcal{K}_{\text{free}}$.

Construction 5.1 (Toric twists Ψ_S and Φ_g)**.** Fix constants $a_g = \frac{1}{10}$, $(\omega_{m,0}, \omega_{m,1}) = (0.015, 0.045)$, $(\omega_{u,0}, \omega_{u,1}) = (0.02, 0.12)$, $v_{\text{comp}} = \frac{1}{4}$, $(\omega_{c,0}, \omega_{c,1}) = (0.05, 0.14)$, $\delta = \frac{1}{4}w_2$, $u_* = -c_s/9$, $c_0 = \delta - c_s/3$, $n_j = -\operatorname{round}(u_j - u_*) \in \mathbb{Z}$, $\phi_j = u_j + n_j - u_*$, and shifted v-targets $(v_1', v_2', v_3') = (-a_g, 0, \frac{1}{2})$. Using $m = 2$ plateau bumps $\beta_v = \beta_{\omega_{m,0},\omega_{m,1},2}$, $\beta_c = \beta_{\omega_{c,0},\omega_{c,1},2}$ (mass $M_c = \omega_{c,0} + \omega_{c,1}$), and $\beta_u = \beta_{\omega_{u,0},\omega_{u,1},2}$:

(1) For $v \in \mathbb{R}$, reduce $\tilde{v} \in [-\frac{1}{3}, \frac{2}{3})$ modulo 1 and set $b_j(v) = \beta_v(\tilde{v} - v_j') - \frac{M_v}{M_c}\beta_c(\tilde{v} - v_{\text{comp}})$, $B_j(v) = \mathcal{B}_v(\tilde{v} - v_j') - \frac{M_v}{M_c}\mathcal{B}_c(\tilde{v} - v_{\text{comp}})$ (so $B_j(v)$ is 1-periodic in $C^5(\mathbb{R})$ because $\int_{-1/3}^{2/3} b_j = 0$). Put $S_v(p,v) = c_0 + P(p)\sum_{j=1}^3 \phi_j b_j(v)$, $S_p(v) = -9\sum_{j=1}^3 \phi_j B_j(v)$, and

$$\Psi_S(G, v, w) = \big(G + S_v(p,v),\ v,\ we^{-2\pi i S_p(v)}\big).$$

(2) For $u \in \mathbb{R}$, reduce $\tilde{u} \in [\frac{1}{5}, \frac{6}{5})$ modulo 1 and set $g_{\text{tw}}(u) = -a_g\big(\beta_u(\tilde{u} - \frac{5}{6}) - \beta_u(\tilde{u} - \frac{7}{12})\big)$ and $G_g(u) = -a_g\big(\mathcal{B}_u(\tilde{u} - \frac{5}{6}) - \mathcal{B}_u(\tilde{u} - \frac{7}{12})\big)$ (1-periodic in $C^4(\mathbb{R})$ and $C^5(\mathbb{R})$). Define

$$\Phi_g(G, v, w) = \big(G,\ v - g_{\text{tw}}(u),\ we^{2\pi i \cdot 9(G_g(u) - u g_{\text{tw}}(u))}\big) \qquad (u = G/P(p)).$$

Proposition 5.2 (Symplecticity, deck equivariance, and marked-fibre routing)**.** *On $\{p < \frac{1}{3}\}$:*

(i) Ψ_S and Φ_g are C^4 symplectomorphisms preserving $p = \pi|w|^2$, with closed-form inverses obtained by reversing the sign of (S_v, S_p) and g_{tw}, and both commute with T_u and T_v.

(ii) Under the window-separation checks (H10) (the four v-intervals $[v_j' \pm 0.045]$, $[0.25 \pm 0.14]$ are pairwise disjoint in $[-\frac{1}{3}, \frac{2}{3})$, and the two u-lobes $[\frac{5}{6} \pm 0.12]$, $[\frac{7}{12} \pm 0.12]$ are disjoint in $[\frac{1}{5}, \frac{6}{5})$ and miss $\frac{1}{3}$ and 1), the preimage of the deck orbit of $\mathcal{M}_j$ under $\Psi = \Phi_g \circ \Psi_S$ is the exact spiral

$$\Psi^{-1}\big(\operatorname{deck}(\mathcal{M}_j)\big) = \big\{(G,v,w) : G \equiv K(p) - \delta \pmod{P(p)},\ v \equiv v_j' \pmod 1\big\}. \tag{12}$$

Since $0 < \delta = \frac{1}{4}w_2 < \frac{1}{2}w_2$ and $3 - 9a > \delta$ (H9b), all three spirals lie inside the free corridor $\mathcal{K}_{\text{free}}$ and are disjoint from $\mathcal{G}_0, \dots, \mathcal{G}_8, F(\mathcal{S}_9), \dots, F(\mathcal{S}_{k-1})$.

Proof. (i) Writing $S(p,v) = c_0 v + P(p)\sum_j \phi_j B_j(v)$ locally, $\partial_v S = S_v$ and $\partial_p S = S_p$, so $d(G + S_v) \wedge dv + dp \wedge d(\theta - S_p) = dG \wedge dv + dp \wedge d\theta$. For Φ_g, writing $H_g(G,p) = P(p)G_g(G/P(p))$ gives $\partial_G H_g = g_{\text{tw}}(u)$ and $\partial_p H_g = -9(G_g(u) - u g_{\text{tw}}(u))$, so Φ_g is the exact time-one flow of H_g (constant along orbits since G and p are preserved) and is C^4 because $g_{\text{tw}} \in C^4$. Deck equivariance holds because b_j, B_j are 1-periodic in v and g_{tw}, G_g are 1-periodic in u (for T_u on Φ_g: shifting $u \mapsto u + 1$ shifts $u g_{\text{tw}}(u)$ by $g_{\text{tw}}(u)$, shifting the fibre phase by $-9g_{\text{tw}}(u)$, which matches T_u's phase $9(v - g_{\text{tw}}(u))$).

(ii) At $u_1 \equiv \frac{5}{6}$, $g_{\text{tw}}(\frac{5}{6}) = -a_g(1 - 0) = -a_g$, whereas at $u_2 \equiv \frac{1}{3}$ and $u_3 \equiv 0$, $g_{\text{tw}} = 0$ by (H10). Thus $\Phi_g^{-1}(\operatorname{deck}(\mathcal{M}_j)) = \{u \equiv u_j \pmod 1,\ v \equiv v_j' \pmod 1\}$. At $v \equiv v_j' \pmod 1$, (H10)

gives $b_j(v_j')=1$, $b_l(v_j')=0$ ($l\neq j$), and $\beta_c(v_j'-v_{\mathrm{comp}})=0$, so $S_v(p,v_j')=c_0+P(p)\phi_j$. Since $\phi_j=u_j+n_j+c_s/9$ and $c_0=\delta-c_s/3$,

$$S_v(p,v_j')=(3-9p)\Big(u_j+n_j+\frac{c_s}{9}\Big)+\delta-\frac{c_s}{3}=P(p)(u_j+n_j)-c_sp+\delta=P(p)(u_j+n_j)-\big(K(p)-\delta\big).$$

Hence $G+S_v(p,v_j')\equiv P(p)u_j \pmod{P(p)}$ if and only if $G\equiv K(p)-\delta \pmod{P(p)}$, proving (12). Since $0<\delta=\frac14 w_2<\min(\frac12 w_2,3-9a)$, in the fundamental domain $[J(p),K(p))$ we have $J(p)+9(a-p)=K(p)-(3-9a)<K(p)-\delta$ (for $p<a$) and $\sup F(\mathcal{S})_G<K(p)-\frac12 w_2<K(p)-\delta$ by (10). □

6. Closed-form transfer to the four-ball

6.1. **Post chart, corner bands, and simplex gluing.** On $\{p<h_K\}$ ($0<h_K<\frac13$, $D=1-3h_K>0$), the post chart $\Pi_{\mathrm{post}}(G,v,w)=(\sigma,\psi,w_{\mathrm{post}})$ with $\psi_0=-\frac16$ and $\ell=1$ sets $u_a=G/P(p)+1$, $\sigma=P(p)v$, $\psi=-\frac16-u_a$, and $\vartheta=\theta-9(u_a-1)v$ ($w_{\mathrm{post}}=\sqrt{p/\pi}\,e^{2\pi i\vartheta}$). Direct differentiation gives $d\sigma\wedge d\psi+dp\wedge d\vartheta=(P\,dv-9v\,dp)\wedge(-P^{-1}dG-9GP^{-2}dp)+dp\wedge(d\theta-9\frac{G}{P}dv-9v\frac{dG}{P}-81\frac{Gv}{P^2}dp)=dG\wedge dv+dp\wedge d\theta$. Under Π_{post}, T_u becomes $\tau_u\colon\psi\mapsto\psi-1$, T_v becomes $\tau_v\colon(\sigma,\psi,p,\vartheta)\mapsto(\sigma+P(p),\psi,p,\vartheta+9\psi+\frac12)$ (because $-9\psi_0=\frac32\equiv\frac12 \pmod 1$), and the three straight marked fibres $(\mathcal{M}_1,\mathcal{M}_2,\mathcal{M}_3)$ become

$$(v_c,\psi)\equiv(0,0),\quad(0,\tfrac12),\quad(\tfrac12,-\tfrac16)\qquad\big(v_c=\sigma/P(p),\ \psi \bmod 1\big).$$

Fix fractions $(f_m,f_{d_2},f_{\sigma_0},f_\delta)=(0.32,0.20,0.20,0.10)$ of D, so $m=0.32D$, $d_2=0.20D$, $\sigma_0=0.20D$, and $\delta_{\mathrm{kh}}=0.10D$, and let K_B be the corner keyhole on $\bar D(h_K)$ with budget $0.8\delta_{\mathrm{kh}}$. Set $(\kappa_0,\kappa_1,\kappa_2)=(-\frac{1}{12},\frac{1}{12},0)$, phase offsets $\mathrm{off}_0=(0,0)$, $\mathrm{off}_1=(\frac{1}{12},\frac14)$, $\mathrm{off}_{2s}=(\frac{1}{12},\frac14)$, $\mathrm{off}_{2c}=(\frac{1}{12},0)$, $\mathrm{left}_0=\mathrm{off}_{2c}$, $\mathrm{left}_1=\mathrm{off}_0$, $\mathrm{left}_2=\mathrm{off}_1$, and $\kappa=e^{-2\pi i/12}=e^{-i\pi/6}$.

Reduce $(\sigma,\psi,p,\vartheta)$ by τ_v^{-n} ($n=\lfloor v_c+\frac16\rfloor$) into the fundamental strip $v_c\in[-\frac16,\frac56)$, and put $c_j=\sigma-j(1-3p)$ for $j\in\{0,1,2\}$. Under the split inequality $2m<D-3\delta_{\mathrm{kh}}$ ($0.64<0.70$), for any $p_2<h_K+\delta_{\mathrm{kh}}$ we have $1-3p_2>D-3\delta_{\mathrm{kh}}>2m$, so the three bands $\{|c_j|<m\}$ ($j=0,1,2$) are pairwise disjoint both before and after the corner keyhole!

(1) *Corner keyhole $\mathcal{B}$ (`Band`).* Outside $\bigcup_{j=0}^2\{|c_j|<m\}$, $\mathcal{B}=\mathrm{id}$. On $\{|c_j|<m\}$, put $\tau_B(c)=1-S_4((|c|-d_2)/(m-d_2))$, $k_j(c)=\kappa_jS_4((c+m)/(2m))$, $\psi_1=\psi+a_j+k_j(c)$, $v_j=\vartheta+b_j+3jk_j(c)-3j\psi_1$ ($(a_j,b_j)=\mathrm{left}_j$), apply $(w_2,g_B)=K_B^{\tau_B(c)}(\sqrt{p/\pi}e^{2\pi iv_j})$, set $\psi_2=\psi_1-\tau_B'(c)g_B$, $p_2=\pi|w_2|^2$, $v_{j,2}=\arg(w_2)/(2\pi)$, recover $\sigma_{\mathrm{out}}=c+j(1-3p_2)$ (preserving c_j!), $\psi_{\mathrm{out}}=\psi_2-a_j-k_j(c)$, $\vartheta_{\mathrm{out}}=v_{j,2}+3j\psi_2-b_j-3jk_j(c)$, and restore τ_v^n.

(2) *Simplex gluing $\mathcal{W}$.* On the output of $\mathcal{B}$ (reduced to $v_c\in[-\frac16,\frac56)$):

- On the four sector pieces outside $\bigcup_{j=0}^2\{|c_j|<m\}$ ($0\le v_c<\frac13$, $\frac13\le v_c<\frac23$, $\frac23\le v_c<\frac56$, and $-\frac16\le v_c<0$, indexed by $s\in\{0,1,2s,2c\}$), set $(\tilde\psi,\tilde\vartheta)=(\psi,\vartheta)+\mathrm{off}_s$ and define $(X_0,X_1,X_2;T_1,T_2)$ by

$$\begin{aligned}
s=0:&\quad(1-2p-\sigma,\ p+\sigma,\ p;\ \tilde\psi,\ \tilde\vartheta-\tilde\psi),\\
s=1:&\quad(p,\ 2-5p-\sigma,\ 4p+\sigma-1;\ -\tilde\vartheta+4\tilde\psi,\ -\tilde\vartheta+5\tilde\psi),\\
s=2s:&\quad(7p+\sigma-2,\ p,\ 3-8p-\sigma;\ \tilde\vartheta-8\tilde\psi,\ -\tilde\psi),\\
s=2c:&\quad(1-2p+\sigma,\ p,\ p-\sigma;\ \tilde\vartheta+\tilde\psi,\ -\tilde\psi),
\end{aligned}$$

  and $W=(\sqrt{X_0},\sqrt{X_1}e^{2\pi iT_1},\sqrt{X_2}e^{2\pi iT_2})\in S^5\subset\mathbb{C}^3$.
- On band $j\in\{0,1,2\}$ ($|c|<m$, $c=c_j$), put $q(c)=\sigma_0(1-S_4(|c|/d_2))$, $S(c)=\sqrt{c^2+q(c)^2}$, $\lambda_+(c)=\frac12(S(c)+c)\ge0$, $\lambda_-(c)=\frac12(S(c)-c)\ge0$ (evaluated without cancellation via $q(c)^2/(2(S(c)+|c|))$), $\lambda(c)=\frac12(c-S(c))=-\lambda_-(c)$, $v_\ell=\mathrm{frac}(\vartheta+b_j+$

$3jk_j(c) - 3j(\psi + a_j + k_j(c))) \in (0,1)$ (since $|c| \le d_2 \implies \tau_B(c) = 1 \implies v_\ell \notin \{0,1\}$ on band interiors where $\lambda' \neq 0, 1$!), $x_a = p + \lambda_+(c)$, $x_b = p + \lambda_-(c)$, $x_r = 1 - x_a - x_b$, $T_a = \psi + a_j + k_j(c) + \lambda'(c)v_\ell$, and permute $(\sqrt{x_r}, \sqrt{x_a}e^{2\pi i T_a}, \sqrt{x_b}e^{2\pi i(v_\ell - T_a)})$ by the cyclic shift $(j, j+1, j+2)$ mod 3.

(3) *Rotated ball chart* χ_κ. Set $Z_0 = (W_2 - \kappa W_1)/\sqrt{2}$, $Z_1 = (W_1 + \bar{\kappa} W_2)/\sqrt{2}$, $Z_2 = W_0$ (unitary rotation in (W_1, W_2), so $|Z|^2 = |W|^2 = 1$). For $Z_0 \neq 0$, set $\chi_\kappa([Z]) = \frac{\bar{Z}_0}{\sqrt{\pi}|Z_0|}(Z_1, Z_2) \in B^\circ(1) \subset \mathbb{C}^2 \cong \mathbb{R}^4$.

Theorem 6.1 (Global regularity, symplecticity, injectivity, and line avoidance of K_{post})**.** *Suppose $0 < h_K < \frac{1}{3}$, K_B satisfies (2) on $\bar{D}(h_K)$ with $\max(\pi d_B^2, \delta_{\mathrm{rel},B}, \delta_{\mathrm{small},B}) \le \delta_{\mathrm{kh}}$ (H13), $0 < d_2 < m$, $2m < D - 3\delta_{\mathrm{kh}}$, $\eta_1 := \frac{2}{3} - (2\delta_{\mathrm{kh}} + S_{\max})/D > 0$, and $m \le 3\eta_1 D$ (H14)–(H15). Then on $\mathcal{U}_{\mathrm{post}} = \{(\sigma, \psi, w_{\mathrm{post}}) : p < h_K\}$:*

(i) Well-definedness and C^7 seams. After $\mathcal{B}$, $p_2 < h_K + \delta_{\mathrm{kh}}$. On every sector piece, $X_0, X_1, X_2 \ge 0$ with $X_0 + X_1 + X_2 = 1$, and only the coordinate not carrying an angle can vanish (at $p = 0$, the angle matches polar coordinates $w_{\mathrm{post}} = \sqrt{p/\pi}e^{2\pi i\vartheta}$ smoothly). On band j, for $|c| \le d_2$ we have $\tau_B(c) = 1$, so $p_2 > x_{0,B} > 0$ and $v_\ell \in (\theta_{0,B}, 1 - \theta_{0,B})$ avoids the branch cut $\{0,1\}$; for $d_2 \le |c| < m$, $q(c) \equiv 0$, so $S(c) = |c|$ ($\lambda'(c) = 0$ for $c \ge d_2$, $\lambda'(c) = 1$ for $c \le -d_2$), making $e^{2\pi i T_a}$ and $e^{2\pi i(v_\ell - T_a)}$ independent of the integer lift of v_ℓ and smooth across $p_2 = 0$! Across all six seams $c_j = \pm m$ (where $\tau_B = 0$, $k_j(-m) = 0$, $k_j(+m) = \kappa_j$) and across the fundamental cut $v_c = -\frac{1}{6} \leftrightarrow \frac{5}{6}$ (via $\mathrm{off}_{2c} - \mathrm{off}_{2s} = (0, -\frac{1}{4})$ and τ_v), the rank-one projector $ZZ^* \in \mathbb{R}^9$ matches identically to order C^7 and is invariant under τ_u, τ_v.

(ii) Exact zero set of Z_0 (line avoidance). For $x \in \mathcal{U}_{\mathrm{post}}$, $Z_0(\mathcal{W}(\mathcal{B}(x))) = 0$ if and only if x lies in the (τ_u, τ_v)-orbit of one of the three straight marked fibres $(0,0)$, $(0, \frac{1}{2})$, $(\frac{1}{2}, -\frac{1}{6})$.

(iii) Symplectic embedding and cancellation-free inverse. On $\mathcal{U}_{\mathrm{post}} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$, $K_{\mathrm{post}} = \chi_\kappa \circ \mathcal{W} \circ \mathcal{B}$ takes values in $B^\circ(1)$, satisfies $K_{\mathrm{post}}^* \omega_0 = d\sigma \wedge d\psi + dw_{\mathrm{post},x} \wedge dw_{\mathrm{post},y}$, and identifies two points if and only if they lie in the same (τ_u, τ_v)-orbit. Its left inverse (`transfer.py`) recovers every sector moment directly from the non-negative coordinate $p = X_2, X_0, X_1$ (never by $1 - X_a - X_b$) and every band moment from $p_2 = X_b - \lambda_-(c)$ ($c \ge 0$) or $X_a - \lambda_+(c)$ ($c < 0$).

Proof. (i) On band j, $x_r = 1 - 2p_2 - S(c) > 1 - 2(h_K + \delta_{\mathrm{kh}}) - S_{\max} = \frac{1}{3} + \frac{2}{3}D - 2\delta_{\mathrm{kh}} - S_{\max} = \frac{1}{3} + \eta_1 D > \frac{1}{3}$ by (H14). On each sector, $\min_l |c_l| \ge m > 0$ ensures both non-p coordinates are $\ge m > 0$ (for instance on sector 1, $c_1 \ge m \implies X_2 = p + c_1 \ge m > 0$ and $c_2 \le -m \implies X_1 = -c_2 + p \ge m > 0$). At $c_0 = -m$: $k_0(-m) = 0$, $\mathrm{left}_0 = \mathrm{off}_{2c}$, $\lambda(-m) = -m$, and $\lambda'(-m) = 1$, giving $(x_r, x_a, x_b) = (1 - 2p + m, p, p + m)$ and $(T_a, v_\ell - T_a) = (\tilde{\vartheta} + \tilde{\psi}, -\tilde{\psi})$, matching sector $2c$; at $c_0 = +m$: $k_0(+m) = \kappa_0 = -\frac{1}{12}$, $\mathrm{left}_0 + (\kappa_0, 0) = (0,0) = \mathrm{off}_0$, $\lambda(+m) = 0$, and $\lambda'(+m) = 0$, giving $(1 - 2p - m, p + m, p)$ and $(\tilde{\psi}, \tilde{\vartheta} - \tilde{\psi})$, matching sector 0. Similarly $c_1 = \pm m$ matches sectors $0 \leftrightarrow 1$ ($\mathrm{left}_1 + (\kappa_1, 3\kappa_1) = \mathrm{off}_1$) and $c_2 = \pm m$ matches sectors $1 \leftrightarrow 2s$ ($\mathrm{left}_2 + (0,0) = \mathrm{off}_{2s}$). Under τ_v, a point near $v_c = -\frac{1}{6}$ in sector $2c$ maps to $v_c' = v_c + 1 \in [\frac{5}{6}, 1)$, where substituting $\sigma' = \sigma + 3 - 9p$ and $\vartheta' = \vartheta + 9\psi + \frac{1}{2}$ into sector $2s$'s formula gives $(X_0', X_1', X_2') = (X_0, X_1, X_2)$, $T_2' = T_2$, and $T_1' = (\vartheta + 9\psi + \frac{1}{2} + \frac{1}{4}) - 8(\psi + \frac{1}{12}) = \vartheta + \psi + \frac{1}{12} = T_1 \pmod 1$.

(ii) Since $|\kappa| = 1$, $Z_0 = 0 \iff W_2 = \kappa W_1$, which requires $X_1 = X_2$ and (when $X_1 = X_2 > 0$) $T_2 - T_1 \equiv -\frac{1}{12} \pmod 1$. Across all seven pieces:

- On band 1 ($X_1 = x_r, X_2 = x_a$), $X_1 - X_2 = 1 - 3p_2 - (\frac{3}{2}S(c) + \frac{1}{2}c) > 0.70D - (\frac{3}{2} \cdot 0.32 + 0.16)D = +0.06D > 0$; symmetrically $X_2 - X_1 > +0.06D > 0$ on band 2.
- On sectors 0, $2c$, $2s$: $X_1 - X_2 = \sigma \ge m > 0$ ($s = 0$), $X_1 - X_2 = \sigma \le -m < 0$ ($s = 2c$), and $X_2 - X_1 = 3 - 9p - \sigma > \frac{1}{6}(3 - 9p) > 0$ ($s = 2s$).

- On band 0 ($X_1 = x_a, X_2 = x_b$): $X_1 = X_2 \iff \lambda_+(c) = \lambda_-(c) \iff c = 0$ ($v_c = 0$). At $c = 0$, $\tau_B(0) = 1$, $\tau_B'(0) = 0$, $\lambda'(0) = \frac{1}{2}$, $k_0(0) = -\frac{1}{24}$, and $\text{left}_0 = (\frac{1}{12}, 0)$, so $T_a = \psi + \frac{1}{24} + \frac{1}{2}v_\ell$ and $T_2 - T_1 = v_\ell - 2T_a = -2\psi - \frac{1}{12}$ (v_ℓ cancels identically), which equals $-\frac{1}{12} \pmod 1$ if and only if $\psi \in \{0, \frac{1}{2}\} \pmod 1$ ($\mathcal{M}_1$ and $\mathcal{M}_2$).
- On sector 1: $X_1 = X_2 \iff 2 - 5p - \sigma = 4p + \sigma - 1 \iff v_c = \frac{1}{2}$ (where $|c_1| = |c_2| = \frac{1}{2}(1 - 3p) > \frac{1}{2}D > m$, outside all bands). There $X_1 = X_2 = \frac{1}{2}(1 - p) > \frac{1}{3}$ and $T_2 - T_1 = \tilde{\psi} = \psi + \frac{1}{12} \equiv -\frac{1}{12} \pmod 1 \iff \psi \equiv -\frac{1}{6} \pmod 1$ ($\mathcal{M}_3$).

(iii) On each sector, $(X_1, X_2) = A_s(p, \sigma) + B_s$ with $A_s \in \mathrm{GL}_2(\mathbb{Z})$ and $(T_1, T_2) = (\tilde{\vartheta}, \tilde{\psi})A_s^{-1}$ gives $\sum_{l=1}^{2} dX_l \wedge dT_l = dp \wedge d\vartheta + d\sigma \wedge d\psi$; on band j, $dx_a \wedge dT_a + dx_b \wedge d(v_\ell - T_a) = dc \wedge d\psi_2 + dp_2 \wedge dv_\ell = d\sigma_{\text{out}} \wedge d\psi_{\text{out}} + dp_2 \wedge d\vartheta_{\text{out}}$, and $\mathcal{B}$ is symplectic by Proposition 3.2. Pulling back ω_0 by χ_κ gives ω_{FS}, whose pullback by W is $\sum_{l=1}^{2} dX_l \wedge dT_l$. Finally, the three band image sets $\mathcal{R}_j = \{|X_{j+1} - X_{j+2}| < m,\ X_j > \frac{1}{3}\}$ are pairwise disjoint: on $\mathcal{R}_0$, if $X_1 = p_2 + \lambda_+(c) > \frac{1}{3}$, then since $p_2 < \frac{1}{3} - \frac{7}{30}D$ and $\max_{|t| \le d_2} \lambda_+(t) = 0.20D < \frac{7}{30}D$, we must have $c > d_2$, so $S(c) = c$, $\lambda_-(c) = 0$, and $X_0 - X_2 = 1 - 3p_2 - c > \frac{7}{10}D - 0.32D = 0.38D > m$; symmetrically $X_0 - X_1 > 0.38D > m$ whenever $X_2 > \frac{1}{3}$. Outside $\mathcal{R}_0 \cup \mathcal{R}_1 \cup \mathcal{R}_2$, $\mathrm{argmin}(X_0, X_1, X_2)$ uniquely identifies sector 0 (X_2 minimal), sector 1 (X_0 minimal), or sector $2s/2c$ ($X_1 = p$ minimal: on $2c$, $X_2 \ge p + m > p$ and $X_0 - X_1 = 1 - 3p + \sigma \ge \frac{1}{2}(1 - 3p) > 0$; since $2c$ is the τ_v^{-1}-image of $v_c' \in [\frac{5}{6}, 1)$ where $2s$'s formula produces the identical W, $2s$'s inverse followed by `reduce` recovers $(v_c, \psi, p, \vartheta)$ exactly). □

7. Explicit parameter verification and proof of Theorem 1.1

7.1. The closed-form parameter tuple $\Pi^*(k, \varepsilon_c; \frac{1}{4})$.

Definition 7.1 (Parameter tuple $\Pi^*(k, \varepsilon_c; f)$)**.** For an integer $k \ge 10$, $\varepsilon_c \in (0, \frac{1}{100}]$, and $f = \frac{1}{4}$, put $s = \sqrt{k}$, $\eta = \varepsilon_c/(1 + \sqrt{1 - \varepsilon_c})$, $t_0 = 1 - \frac{1}{2}\eta$, $a = t_0/s$, $h = t_0/3$, $c_s = 3s - 9$, $g = \frac{3}{2}\eta$, $T = g/c_s$, $(\lambda_a, \lambda_2, \mu) = (\frac{9}{10}, \frac{3}{10}, \frac{1}{2})$, $w_G = \lambda_a g$, $w_2 = \lambda_2 g$, $\upsilon = \sqrt{11/1000}$.

(1) *Comparator keyhole K_C:* $r_a = \sqrt{a/\pi}$, $d = \sqrt{fT/\pi}/(1+\upsilon)$, $\beta = fT/\big(\pi(r_a+d)(r_a+8.4d)\big)$, $W_1 = \beta/(3(1+\beta))$, $r_h = 0.9d\sin(2\pi W_1)$, $\rho_1 = 0.8((d - r_h)/(d + r_a))^2$, $W_2 = \beta/(1 + \beta - \rho_1) - W_1$, $X_w = 0.2\pi(d - r_h)^2$.

(2) *Post height h_K and $D = 1 - 3h_K$:* $h_K = \max\big(a,\ O(-w_G) - \frac{1}{2}\mu T(1 - \lambda_a),\ h - \frac{1}{2}\mu T\big)$, so $D = \min\big(\frac{s-3+\frac{3}{2}\eta}{s},\ \frac{s-3+w_G}{s} + \frac{3}{2}\mu T(1 - \lambda_a),\ \frac{1}{2}\eta + \frac{3}{2}\mu T\big) > 0$.

(3) *Corner keyhole K_B:* $(m, d_2, \sigma_0, \delta_{\text{kh}}) = (0.32, 0.20, 0.20, 0.10)D$, $r_B = \sqrt{h_K/\pi}$, $d_B = \sqrt{0.8\delta_{\text{kh}}/\pi}/1.05$, $\beta_B = 0.8\delta_{\text{kh}}/\big(\pi(r_B + d_B)(r_B + 8.4d_B)\big)$, with $(W_{1,B}, r_{h,B}, \rho_{1,B}, W_{2,B}, X_{w,B})$ by the same 'keyhole_star' formulas with ρ_1-factor 0.7.

Lemma 7.2 (Grid-free analytic bounds for (H6) and (H7))**.** *With $(\lambda_a, \lambda_2, \mu, f) = (\frac{9}{10}, \frac{3}{10}, \frac{1}{2}, \frac{1}{4})$, suppose $\delta_{\text{rel}} = fT = \frac{1}{4}T$ and $\delta_{\text{small}} \le 0.04T$. Then (H6) and (H7) hold on all of $[0, 1]$ with margin $> 0.012T$.*

Proof. Divide by T. For (H6), $(1-\mu)(1-\lambda_a+u(\lambda_a-\lambda_2)) = 0.05+0.30u \ge 0.05 > 0.04 \ge \delta_{\text{small}}/T$. It remains to bound $h_6(u) = 0.05 + 0.30u - 0.25S_4(u)$ on $[0, 1]$. On $[0, \frac{1}{2}]$, Lemma 2.1(i) gives $S_4(u) \le u$, so $h_6(u) \ge 0.05+0.05u \ge 0.05$. On $[\frac{1}{2}, 1]$, S_4 is concave, so for $u_0 = \frac{2}{3}$ ($S_4(\frac{2}{3}) < 0.9414$, $S_4'(\frac{2}{3}) = \frac{256}{70}\sin^8(\frac{2\pi}{3}) = \frac{81}{70} < 1.1572$), $S_4(u) \le S_4(u_0) + S_4'(u_0)(u - u_0)$, bounding $h_6(u)$ from below by an affine function whose values at $u = \frac{1}{2}$ and $u = 1$ are > 0.0128 and > 0.0182. For (H7), since $\max(\delta_{\text{rel}}, \delta_{\text{small}})/T = \frac{1}{4}$, we must bound $h_7(u) = 0.5(0.7 + 0.3u) - 0.25 - 0.3I_4(u) = 0.10 + 0.15u - 0.30I_4(u)$ on $[0, 1]$. Its derivative $0.30(\frac{1}{2} - S_4(u))$ is ≥ 0 on $[0, \frac{1}{2}]$ and ≤ 0 on $[\frac{1}{2}, 1]$, so $\min_{[0,1]} h_7 = \min(h_7(0), h_7(1)) = 0.10 > 0$. □

Proposition 7.3 (Universal validity of (H0)–(H15))**.** *For every integer $k \geq 10$ and every $\varepsilon_c \in (0, \frac{1}{100}]$, $\Pi^*(k, \varepsilon_c; \frac{1}{4})$ satisfies all hypotheses of Theorems 3.3, 4.2, 5.2, and 6.1, as well as $r_+ > 1$.*

Proof. Since $s \geq \sqrt{10} > 3.16227$ and $0 < \eta \leq 1 - \sqrt{0.99} < 0.005013$, we have $t_0 > 0.9974$, $s/(s-3) \leq \sqrt{10}/(\sqrt{10}-3) = 10 + 3\sqrt{10} < 19.49$, and

$$\frac{T}{a} = \frac{\frac{1}{2}\eta/(s-3)}{t_0/s} = \frac{\eta}{2t_0}\,\frac{s}{s-3} < \frac{0.005013}{1.9948} \cdot 19.49 < 0.049.$$

(H2) and $r_+ > 1$: Since $0 < m_s \leq \frac{1}{8}\eta$ and $\eta < 0.01$, $(1-m_s)^2 > 1 - 2m_s \geq 1 - \frac{1}{4}\eta$, while $(1-\frac{1}{2}\eta)/(1-\eta) = 1 + \frac{\eta}{2(1-\eta)} > 1 + \frac{1}{2}\eta$. Thus $r_+^2 = (1-m_s)^2(1-\frac{1}{2}\eta)/(1-\eta) > (1-\frac{1}{4}\eta)(1+\frac{1}{2}\eta) = 1 + \frac{1}{4}\eta(1-\frac{1}{2}\eta) > 1$, which is equivalent to $\rho_b = \rho_{\mathrm{ball}}/(1-m_s)^2 < a$.

(H3)–(H5) for K_C: Since $\pi(r_a + d)(r_a + 8.4d) > \pi r_a^2 = a$, $\beta < \frac{1}{4}(T/a) < 0.0123$. Also $\pi d^2 = \frac{1}{4}T/(1+\upsilon)^2 < 0.205T$, so $(d/r_a)^2 < 0.205(0.049) < 0.0101$ ($d/r_a < 0.101$). By definition, (K1) and (K2) hold with equality/$0.9 < 1$, $r_h < 0.9d(2\pi W_1) < 0.024d < d$, $\rho_1 < 0.8(0.101/1.101)^2 < 0.0068 < 1$, $W_2 - W_1 = \beta\big(\frac{1}{1+\beta-\rho_1} - \frac{2}{3(1+\beta)}\big) \geq \frac{\beta}{3(1+\beta)} > 0$, and $W_2 < W_1 + W_2 = \beta/(1+\beta-\rho_1) < 0.0123/0.993 < 0.013 < \frac{1}{6}$. For (K3), $\rho_1\pi(d+r_a)^2 + \frac{1}{2}X_w = (0.8+0.1)\pi(d-r_h)^2 = 0.9\pi(d-r_h)^2 < \pi(d-r_h)^2$ holds identically! Moreover, $d/(r_a + 8.4d) < \frac{1}{8.4}$ and $W_2/\beta < \frac{1}{0.993} - \frac{1}{3.04} < 0.68$, so

$$\frac{\delta_{\mathrm{small}}}{fT} = 4\pi^2\,\frac{d}{r_a + 8.4d}\Big(\frac{W_2}{\beta}\Big)^2\beta < \frac{4\pi^2}{8.4} \cdot (0.68)^2 \cdot 0.0123 < 0.027 < 0.04.$$

Lemma 7.2 then gives (H6) and (H7).

(H8), (H9), (H9b): $\pi d^2 < 0.205T < 0.35T = T(1-\mu-\frac{1}{2}\lambda_2)$, and $s(\pi d^2 + 0.65T) < 0.855sT = 0.4275\eta\frac{s}{s-3} < 0.042 < 1 - 0.45\eta = sO(-w_G)$, giving (H8). For (H9), $c_s\pi r_h^2 < c_s\pi(0.024d)^2 < 6 \cdot 10^{-4}c_s(0.205T) < 2 \cdot 10^{-4}g$, so $c_s\pi r_h^2 + 1.5\lambda_2 g < (0.45 + 0.001)g < g$. For (H9b), $a < 1/\sqrt{10} \implies 3 - 9a > 3(1 - 9/\sqrt{10}) > 0.1539$, while $\delta = \frac{1}{4}w_2 = \frac{9}{80}\eta < 0.0006 < 0.1539$. (H10) is a check of fixed rational constants ($0.045 + 0.045 < 0.10$, $0.045 + 0.14 < 0.25$, $2 \cdot 0.12 < \frac{1}{4}$, $\frac{1}{6} - 0.12 - 0.02 > 0.02 > 0$).

(H11)–(H15): By Definition 7.1(2), $D > 0$ ($h_K < \frac{1}{3}$), $h_K \geq a$, and since $O(G) - \mu\,\mathrm{slack}(G)$ is piecewise affine on $[-w_G, 0] \cup [0, G_{\max}]$, its maximum is bounded by $\max\big(O(-w_G) - \mu T(1-\lambda_a), O(0) - \mu T, O(G_{\max}) - \mu T\big)$; note $O(0) < O(G_{\max}) = h$ when $G_{\max} > 0$ ($O(-w_G)$ otherwise), and subtracting $\frac{1}{2}\mu T(1-\lambda_a) > 0$ in h_K ensures strict inequality $h_K > O(G) - \mu\,\mathrm{slack}(G)$ everywhere on $[-w_G, G_{\max}]$. Finally, $h_K \geq a > \frac{0.997}{s}$ while $\delta_{\mathrm{kh}} = 0.1D$: since $D = 1 - 3h_K \leq 1 - 3a < 1$ and even as $s \to \infty$, $h_K \geq \max(a, \frac{1}{3} - \frac{1}{3}(\frac{1}{2}\eta + \frac{3}{2}\mu T)) > \frac{1}{3} - 0.01 > 0.32$, we always have $h_K > 0.31$ (for $k = 10$, $a > 0.315$; for $k \geq 11$, $O(-w_G)$ or $h = \frac{1}{3}t_0 > 0.332$ gives $h_K > 0.315$!). Thus $\delta_{\mathrm{kh}}/h_K < 0.10/0.315 < 0.32$ and $\beta_B < 0.8(\delta_{\mathrm{kh}}/h_K)/(1) < 0.003$ (since $D \leq \frac{1}{2}\eta + \frac{3}{2}\mu T < 0.015 \implies \delta_{\mathrm{kh}} < 0.0015$). Exactly as for K_C, K_B satisfies (K1)–(K3) with $\rho_{1,B}\pi(d_B + r_B)^2 + \frac{1}{2}X_{w,B} = 0.8\pi(d_B - r_{h,B})^2$, $\delta_{\mathrm{rel},B} = 0.8\delta_{\mathrm{kh}}$, $\pi d_B^2 = 0.8\delta_{\mathrm{kh}}/1.05^2 < \delta_{\mathrm{kh}}$, and $\delta_{\mathrm{small},B} < 0.01\delta_{\mathrm{kh}}$, proving (H12)–(H13). On $[-m, m]$, for $|c| \leq d_2$ we have $S(c) \leq \sqrt{d_2^2 + \sigma_0^2} = 0.20\sqrt{2}\,D < 0.283D < 0.32D = m$, and for $d_2 \leq |c| \leq m$, $S(c) = |c| \leq m$; thus $S_{\max} = 0.32D$, $\eta_1 = \frac{2}{3} - (0.20 + 0.32) = \frac{11}{75} > 0.1466 > 0$, $3\eta_1 = \frac{11}{25} = 0.44 \geq 0.32 = m/D$ (H14–H15), and $2m/D + 3\delta_{\mathrm{kh}}/D = 0.64 + 0.30 = 0.94 < 1$ ('split'). □

7.2. **Completion of the proof of Theorem 1.1.**

Proof of Theorem 1.1. Fix $k \geq 10$ and $\varepsilon \in (0, 1)$. Since $\varepsilon_c = \min(\varepsilon, \frac{1}{100}) \leq \varepsilon$, $R_c \leq R_{\mathrm{req}}$, and Proposition 7.3 gives $r_+ > 1$ and all hypotheses (H0)–(H15).

(1) Let $X = (q_1, q_2, p_1, p_2) \in \mathbb{B}^\circ(r_+)$. Stage 1, $S_{\mathrm{in}}(X) = (q_1, p_1, q_2, p_2)/(\sqrt{\pi}R_s)$, satisfies $\pi|S_{\mathrm{in}}(X)|^2 = |X|^2/R_s^2 < r_+^2\rho_b = a$ and pulls back $dx \wedge dy + \omega_V$ to $(\pi R_s^2)^{-1}\omega_0$.

(2) By Proposition 2.2, KR maps $B^\circ(a)$ real-analytically, bijectively, and symplectically onto $\mathcal{C}_a$.
(3) By Lemma 4.1 and Theorem 4.2, $F^{(i)} \circ C_i$ $(0 \le i < k)$ are C^7 symplectic embeddings on $\mathcal{C}_a$ whose images $U_i = F^{(i)}(C_i(\mathcal{C}_a))$ are pairwise disjoint modulo $\langle T_u, T_v\rangle$, lie in $\{p < h_K\}$, and miss the corridor $\mathcal{K}_{\mathrm{free}}$.
(4) By Proposition 5.2, $\Psi = \Phi_g \circ \Psi_S$ is a C^4 deck-equivariant symplectomorphism preserving $\{p < h_K\}$ and mapping $\Psi(U_i)$ into $\{p < h_K\} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$, keeping $\Psi(U_0), \dots, \Psi(U_{k-1})$ pairwise disjoint modulo $\langle T_u, T_v\rangle$.
(5) By Theorem 6.1, $K_{\mathrm{post}} \circ \Pi_{\mathrm{post}}$ is C^7, symplectic, deck-invariant, and injective modulo $\langle T_u, T_v\rangle$ on $\{p < h_K\} \setminus \mathrm{deck}\{\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3\}$ into $B^\circ(1)$. Hence $\tilde{f}_i = K_{\mathrm{post}} \circ \Pi_{\mathrm{post}} \circ \Psi \circ F^{(i)} \circ C_i \circ \mathrm{KR}$ are C^4 symplectic embeddings on $B^\circ(a)$ with pairwise disjoint images in $B^\circ(1)$.
(6) Finally, $S_{\mathrm{out}}(y_1, y_2, y_3, y_4) = \sqrt{\pi} R_s(y_1, y_2, y_3, y_4)$ maps $B^\circ(1)$ onto $\mathbb{B}^\circ(R_s) \subset \mathbb{B}^\circ(R_{\mathrm{req}})$ with conformal factor πR_s^2, cancelling Stage 1's $(\pi R_s^2)^{-1}$ so that $f_i^* \omega_0 = \omega_0$.
(7) For $Y \in \mathbb{R}^4$, Preimage(j, Y) checks $|Y| < R_s$ (via $1 - \pi|Y/(\sqrt{\pi} R_s)|^2 > 0$ in 'unchart'), applies to_fund $\circ\, \Psi^{-1} \circ \Pi_{\mathrm{post}}^{-1} \circ \mathcal{B}^{-1} \circ (\chi_\kappa \circ \mathcal{W})^{-1}$ to obtain the unique fundamental representative $y_{\mathrm{fl}} \in \mathcal{F}_{\mathrm{fl}}$, applies $(F^{(j)})^{-1}$ and C_j^{-1} to obtain $(X_{\mathrm{cell}}, T_{\mathrm{cell}}, w)$, tests membership in $\mathcal{C}_a$ $(p < a,\ 0 < X_{\mathrm{cell}} < a - p,\ 0 < T_{\mathrm{cell}} < 1)$, and applies $S_{\mathrm{in}}^{-1} \circ \mathrm{KR}^{-1}$. When $Y = f_i(X)$ with $X \in \mathbb{B}^\circ(r_+)$, deck invariance of $K_{\mathrm{post}} \circ \Pi_{\mathrm{post}}$ and Theorem 4.2(ii) guarantee that to_fund recovers the exact point $(F^{(i)} \circ C_i \circ \mathrm{KR} \circ S_{\mathrm{in}})(X) \in \mathcal{F}_{\mathrm{fl}}$, giving Preimage$(i, f_i(X)) = (X, \mathrm{true})$ and disjointness for $j \neq i$. □

Remark 7.4 (Exact arithmetic versus IEEE-754 `float64`). Theorem 1.1 holds over $\mathbb{R}$ for all $k \ge 10$ and $\varepsilon \in (0, 1)$. In `float64`, `solution(k, eps)` evaluates the analytic margins of Proposition 7.3 on the rounded parameters and raises `ValueError` if machine precision underflows a strict inequality (for instance $\varepsilon \le 10^{-16}$, where `1.0 - eps == 1.0` makes $a - \rho_b = 0$). Within the `float64`-admissible range, all three scalar Newton solvers run 60 safeguarded iterations on bounded brackets (reaching 53-bit accuracy in ≤ 8 steps once quadratic), and the inverse transfer recovers every moment coordinate without subtraction cancellation. Because gaps of size $O(\varepsilon_c/\sqrt{k})$ sit next to $O(1)$ coordinates, local Lipschitz constants reach 10^8–10^{15} for small ε or large k; arbitrary-precision evaluation via `program/check_embedding_mp.py` verifies the submitted Python source directly to 100+ digits.

Authorship and the role of AI

Liang Song is the sole author and is responsible for the manuscript and its release. The mathematics, the code and the text were produced by AI systems, including GPT-6.1 Sol (OpenAI), Claude Opus 5.5 (Anthropic), Gemini (Google) and other models, working in a research environment designed and directed by Liang Song, who chose the problem, assembled the context, designed the verification pipeline and made the editorial decisions. The theorems of this paper are not part of the Lean 4 formalisation of Papers I–II; numerical and arbitrary-precision checks are documented in `program/README.md`.

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